

LANDSLIDES, AVALANCHES AND GLOF RISK

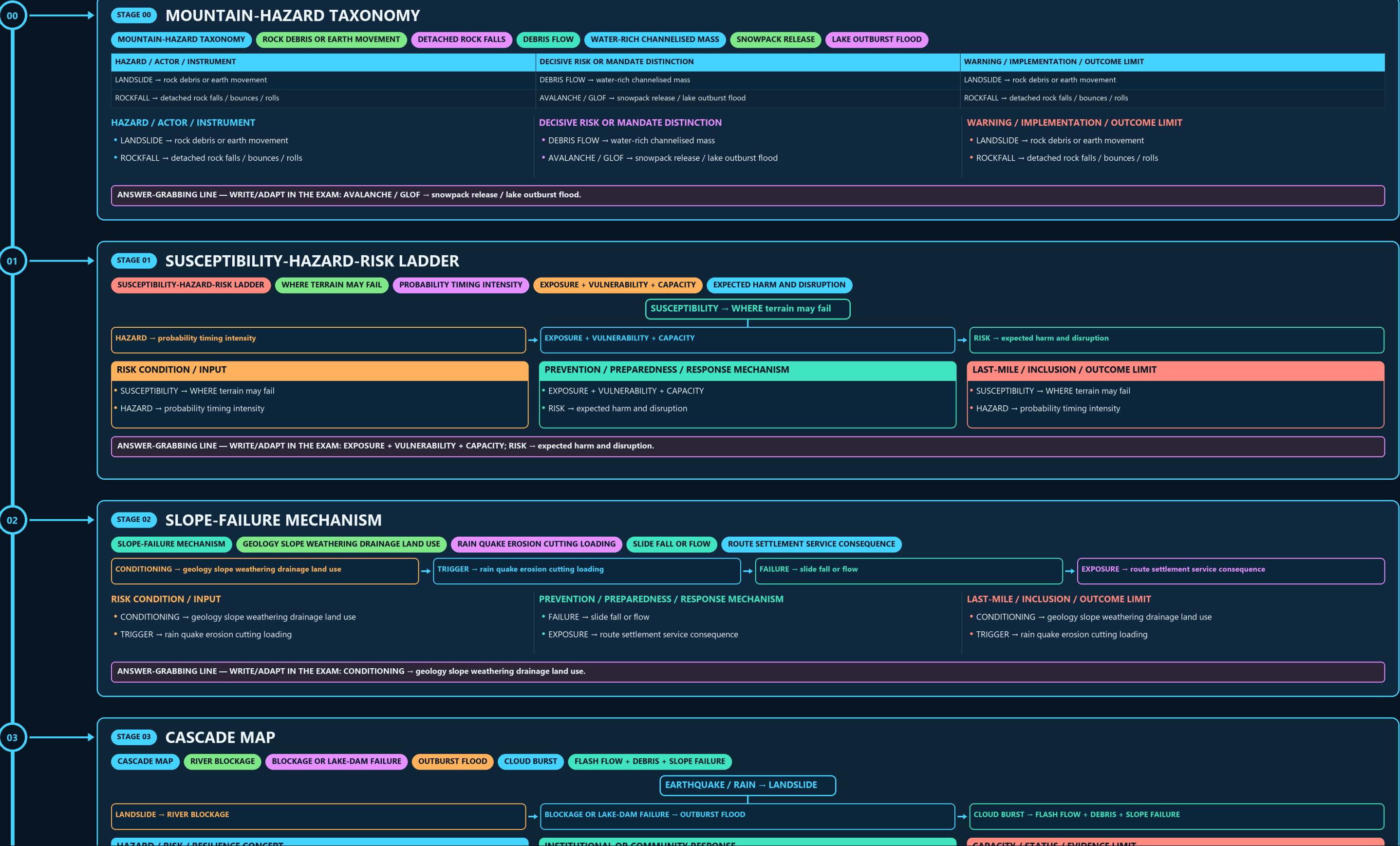
MOUNTAIN-HAZARD TAXONOMY → SUSCEPTIBILITY-HAZARD-RISK LADDER → SLOPE-FAILURE MECHANISM → CASCADE MAP → INSTITUTIONAL OWNER MATRIX → GLACIAL-LAKE MONITORING RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



01

STAGE 01

SUSCEPTIBILITY-HAZARD-RISK LADDER

SUSCEPTIBILITY-HAZARD-RISK LADDER

WHERE TERRAIN MAY FAIL

PROBABILITY TIMING INTENSITY

EXPOSURE + VULNERABILITY + CAPACITY

EXPECTED HARM AND DISRUPTION

HAZARD → probability timing intensity

→

EXPOSURE + VULNERABILITY + CAPACITY

→

RISK → expected harm and disruption

RISK CONDITION / INPUT

• SUSCEPTIBILITY → WHERE terrain may fail

• HAZARD → probability timing intensity

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

• EXPOSURE + VULNERABILITY + CAPACITY

• RISK → expected harm and disruption

LAST-MILE / INCLUSION / OUTCOME LIMIT

• SUSCEPTIBILITY → WHERE terrain may fail

• HAZARD → probability timing intensity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPOSURE + VULNERABILITY + CAPACITY; RISK → expected harm and disruption.

02

STAGE 02

SLOPE-FAILURE MECHANISM

SLOPE-FAILURE MECHANISM

GEOLOGY SLOPE WEATHERING DRAINAGE LAND USE

RAIN QUAKE EROSION CUTTING LOADING

SLIDE FALL OR FLOW

ROUTE SETTLEMENT SERVICE CONSEQUENCE

CONDITIONING → geology slope weathering drainage land use

→

TRIGGER → rain quake erosion cutting loading

→

FAILURE → slide fall or flow

→

EXPOSURE → route settlement service consequence

RISK CONDITION / INPUT

• CONDITIONING → geology slope weathering drainage land use

• TRIGGER → rain quake erosion cutting loading

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

• FAILURE → slide fall or flow

• EXPOSURE → route settlement service consequence

LAST-MILE / INCLUSION / OUTCOME LIMIT

• CONDITIONING → geology slope weathering drainage land use

• TRIGGER → rain quake erosion cutting loading

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONDITIONING → geology slope weathering drainage land use.

03

STAGE 03

CASCADE MAP

CASCADE MAP

RIVER BLOCKAGE

BLOCKAGE OR LAKE-DAM FAILURE

OUTBURST FLOOD

CLOUD BURST

FLASH FLOW + DEBRIS + SLOPE FAILURE

LANDSLIDE → RIVER BLOCKAGE

→

BLOCKAGE OR LAKE-DAM FAILURE → OUTBURST FLOOD

→

CLOUD BURST → FLASH FLOW + DEBRIS + SLOPE FAILURE

HAZARD / RISK / RESILIENCE CONCERN

EARTHQUAKE / RAIN → LANDSLIDE

INSTITUTIONAL OR COMMUNITY RESPONSE

CAPACITY / STATUS / EVIDENCE LIMIT

- **CONDITIONING** → geology slope weathering drainage land use
- **TRIGGER** → rain quake erosion cutting loading

- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER.

06

STAGE 06 ZONATION-TO-DECISION BRIDGE

ZONATION-TO-DECISION BRIDGE

HIGHER SUSCEPTIBILITY

AVOID OR REQUIRE SITE INVESTIGATION

CONTROL LAND USE DRAINAGE CUTTING LOADING

MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HIGHER SUSCEPTIBILITY / HAZARD?

AVOID OR REQUIRE SITE INVESTIGATION

CONTROL LAND USE DRAINAGE CUTTING LOADING

MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HAZARD / RISK / RESILIENCE CONCEPT

- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

INSTITUTIONAL OR COMMUNITY RESPONSE

- CONTROL LAND USE DRAINAGE CUTTING LOADING
- MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

CAPACITY / STATUS / EVIDENCE LIMIT

- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTROL LAND USE DRAINAGE CUTTING LOADING; MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK.

07

STAGE 07 SLOPE-MEASURE PORTFOLIO

SLOPE-MEASURE PORTFOLIO

SUBSURFACE DRAINAGE

CONTROLLED EXCAVATION AND TOE PROTECTION

RETAINING PROTECTION BIOENGINEERING

SITE-SPECIFIC DESIGN MAINTENANCE RESIDUAL RISK

WATER → surface / subsurface drainage

GEOMETRY → controlled excavation and toe protection

MATERIAL → retaining protection bioengineering

LIMIT → site-specific design maintenance residual risk

HAZARD / RISK / RESILIENCE CONCEPT

- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

INSTITUTIONAL OR COMMUNITY RESPONSE

- MATERIAL → retaining protection bioengineering
- LIMIT → site-specific design maintenance residual risk

CAPACITY / STATUS / EVIDENCE LIMIT

- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MATERIAL → retaining protection bioengineering; LIMIT → site-specific design maintenance residual risk.

08

STAGE 08 ROUTE AND SETTLEMENT RESILIENCE

ROUTE AND SETTLEMENT RESILIENCE

SITING + SPOIL

DRAINAGE CONTROL

ALTERNATIVE ACCESS + SAFE STOPPAGE

ASSEMBLY AREA + EVACUATION ROUTE

HOSPITAL POWER TELECOM WATER CONTINUITY

ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL

ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ASSEMBLY AREA + EVACUATION ROUTE

HOSPITAL POWER TELECOM WATER CONTINUITY

HAZARD / RISK / RESILIENCE CONCEPT

- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

INSTITUTIONAL OR COMMUNITY RESPONSE

- ASSEMBLY AREA + EVACUATION ROUTE
- HOSPITAL POWER TELECOM WATER CONTINUITY

CAPACITY / STATUS / EVIDENCE LIMIT

- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL.

09

STAGE 09 WARNING-TO-ACTION CHAIN

WARNING-TO-ACTION CHAIN

COMMUNICATE UNCERTAINTY

CLOSE ROUTE

SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

MONITOR → ASSESS → FORECAST / BULLETIN

AUTHORISE → COMMUNICATE UNCERTAINTY

CLOSE ROUTE / EVACUATE → SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

RISK CONDITION / INPUT

- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- CLOSE ROUTE / EVACUATE → SAFE DESTINATION
- VERIFY RECEIPT ACTION AND FEEDBACK

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL.

09

STAGE 09 WARNING-TO-ACTION CHAIN

WARNING-TO-ACTION CHAIN COMMUNICATE UNCERTAINTY CLOSE ROUTE SAFE DESTINATION VERIFY RECEIPT ACTION AND FEEDBACK

MONITOR → ASSESS → FORECAST / BULLETIN

AUTHORISE → COMMUNICATE UNCERTAINTY

CLOSE ROUTE / EVACUATE → SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

RISK CONDITION / INPUT

- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- CLOSE ROUTE / EVACUATE → SAFE DESTINATION
- VERIFY RECEIPT ACTION AND FEEDBACK

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE ROUTE / EVACUATE → SAFE DESTINATION; VERIFY RECEIPT ACTION AND FEEDBACK.

10

STAGE 10 DEVELOPMENT-RISK BALANCE

DEVELOPMENT-RISK BALANCE CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK RISK-INFORMED SITING NOT BLANKET CLAIMS

HAZARD / ACTOR / INSTRUMENT

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

DECISIVE RISK OR MANDATE DISTINCTION

VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

WARNING / IMPLEMENTATION / OUTCOME LIMIT

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS

HAZARD / ACTOR / INSTRUMENT

- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

DECISIVE RISK OR MANDATE DISTINCTION

- VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS
- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK
- VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK; VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS.

11

STAGE 11 MOUNTAIN-RISK ANSWER SPINE

MOUNTAIN-RISK ANSWER SPINE CLASSIFY PROCESS SEPARATE CONDITION FROM TRIGGER DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER

DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION

COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

HAZARD / RISK / RESILIENCE CONCEPT

- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER
- DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

INSTITUTIONAL OR COMMUNITY RESPONSE

- ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION
- COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

CAPACITY / STATUS / EVIDENCE LIMIT

- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER
- DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT NDMA'S EXPLICIT NON-ISSUANCE OF CLOUDBURST-SPECIFIC GUIDELINES STATE-LEVEL RESPONSIBILITY FOR AREA-SPECIFIC MEASURES AVALANCHE WARNING SITS WITH A DEFENCE-RESEARCH INSTITUTION, NOT MINING-LINKED LANDSLIDES AS A DISTINCT SUB-CATEGORY

THE UTTARAKHAND 2013 DISASTER AS A DOCUMENTED MULTI-CLOUDBURST

CLOUDBURST'S NEAR-IRREDUCIBLE PREDICTION GAP

VISIONIAS STATES

OPTIONAL RISK / RESILIENCE DEPTH

- NDMA's explicit non-issuance of cloudburst-specific guidelines
- State-level responsibility for area-specific measures : landslide
- CAUTION: Avalanche warning sits with a defence-research institution, not
- Mining-linked landslides as a distinct sub-category : VisionIAS's

MANDATE, WARNING OR OUTCOME LIMIT

- The Uttarakhand 2013 disaster as a documented multi-cloudburst,
- CAUTION: Cloudburst's near-irreducible prediction gap : VisionIAS states
- ISRO's Himalayan glacial-lake monitoring role (PDF p. 79) is the
- CAUTION: Avalanche prediction shares cloudburst's short-lead-time

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- CAUTION: Himalayan/hill-state populations dependent on tourism and
- CAUTION: Hydropower/infrastructure development versus glacial-risk
- CAUTION: Terrace farming versus Jhumming (shifting cultivation) in the
- NOT: The "landslide vulnerability zones are mapped, therefore risk is

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.